

Subrahmanya Rajesh Nayak

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EDUCATION

M.Sc. in Big Data and Artificial Intelligence

2024 - Present

SRH Berlin University of Applied Sciences

Grade: 1.5

B.E. in Information Science and Engineering

2020 - 2024

NMAM Institute of Technology, Karkala, India

CGPA: 8.37/10

Technical Skills

Programming Languages: Python, Java, R, SQL, C

Machine Learning & Data Science: TensorFlow, PyTorch, Scikit-learn, Keras, Pandas, NumPy, Hugging Face Transformers

Model Development & MLOps: TensorBoard, MLflow, Kubeflow, Airflow, Git, Docker, REST APIs

Big Data & Cloud Platforms: Apache Spark, Hadoop, Hive, Kafka, AWS, GCP, Azure

Databases: MySQL, MongoDB

Tools & Other Skills: Tableau, Power BI, Linux, Shell Scripting, Agile Methodologies, Data Visualization, NLP, Deep Learning

EXPERIENCE

Research Intern

Jan 2024 – Jun 2024

NITK Centre, Surathkal

Cybersecurity using Deep Learning (CAN Intrusion Detection in Automated Vehicles)

- Engineered an anomaly detection system using **Python**, **TensorFlow**, and **Keras**.
- Applied **LSTM** models to benchmark datasets for detecting CAN bus intrusions.
- Enhanced model accuracy via robust preprocessing and hyperparameter tuning.
- Delivered biweekly technical presentations to faculty supervisors.

Full Stack Developer Intern

Aug 2022 – Sep 2022

Zephyr Technologies, Remote

Online Quiz Platform

- Built a responsive full-stack quiz platform using **React**, **HTML/CSS**, **JavaScript**, and **Firebase**.
- Implemented dynamic quiz logic, timers, scoring, and secure user authentication.
- Integrated **Firebase Authentication** and **Realtime Database** for real-time performance.

Web and Mobile App Developer Intern

May 2022 – Jun 2022

RineX, Remote

Yoga Pose Guidance App

- Developed a yoga mobile app using **Android Studio** and **Java**.
- Implemented guided pose visuals, audio instructions, and timers.
- Focused on delivering a seamless user experience through intuitive UI/UX.

Projects

LLM-Powered Chatbot with RAG on Custom Domain

LangChain, Hugging Face Transformers, FAISS, Streamlit, Python

- Built a domain-specific chatbot using Retrieval-Augmented Generation (RAG) to answer contextual queries.
- Embedded documents using sentence-transformers and implemented semantic search via FAISS.
- Integrated Hugging Face LLMs to generate responses and deployed the app using Streamlit.

LLM-Based Codebase Q&A Assistant with RAG

LangChain, FAISS, Tree-sitter, LLaMA (Ollama), Python

- Developed an AI assistant that understands and answers natural language queries over complex codebases.
- Parsed code with AST tools and generated embeddings for semantic chunk retrieval using FAISS.
- Used a local LLM (LLaMA 3) for low-latency offline inference simulating Copilot-like functionality.

Fine-Tuned Transformer for Multilingual Sentiment Analysis

Hugging Face Transformers, XLM-RoBERTa, PyTorch, Google Colab

- Fine-tuned a multilingual transformer model for sentiment classification across English, German, and Italian.
- Preprocessed datasets, trained on Google Colab GPU, and evaluated performance using F1-score and confusion matrix.
- Published model on Hugging Face Model Hub for reproducibility and community use.

Deep Learning for Cataract Detection Using Fundus and Naked Eye Imaging

Python, TensorFlow, Keras, OpenCV, CNNs, DCNNs, Flask

- Designed CNN-based models to detect cataracts from fundus and real-time naked eye images.
- Used data augmentation and preprocessing techniques to improve model robustness.
- Achieved high accuracy on benchmark datasets and deployed a prototype for real-world testing.

AQI Live Monitoring and Prediction

Python, Flask, Scikit-learn, HTML, CSS, JavaScript

- Developed a web dashboard for real-time AQI visualization using environmental sensor data.
- Trained regression models to predict AQI based on weather parameters and pollution indices.
- Deployed using Flask with dynamic data streaming and user-friendly UI.

PUBLICATIONS

- “Deep Learning Approaches for Cataract Detection Using Fundus and Real-Time Naked Eye Imaging”
Conference-2024 International Conference on Recent Advances in Science and

LANGUAGES

English (Fluent), German (A2)

Hobbies

Problem solving, Traveling, Music production, Singing